

Linear independence

If V is a vector space we say that $v_1, \dots, v_n \in V$ are linearly independent if whenever

$$c_1 v_1 + \dots + c_n v_n = 0$$

then $c_1, \dots, c_n$ are necessarily 0. A set of vectors B in V is linearly independent if every finite subset of B is linearly independent.

Ex: ① $V = P_n$, all poly. of $\deg \leq n$.

$1, x, x^2, \dots, x^n$ is lin. indep.

② $V = P$, all polynomials.

$1, x, x^2, x^3, x^4, \dots$ is lin. indep.

Any finite subset is lin. indep by ①.

A basis for a vector space

If B is a subset of a vector space V then B is a basis for V if

- 1 B is linearly independent, and
- 2 B spans V .

Ex: ① $e_1, \dots, e_n$ in $\mathbb{R}^n$ is a basis for $\mathbb{R}^n$
(standard basis for $\mathbb{R}^n$).

② $1, x, x^2, \dots, x^n$ is a basis for P_n .

ii) $1, x, x^2, \dots$ is a basis for P . $\leftarrow P$ has
an infinite basis.

Ex: In $\mathbb{R}^3$, $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ is a basis

for $\mathbb{R}^3$. Why?

$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \leftarrow$ invertible
since its
det is $1 (\neq 0)$

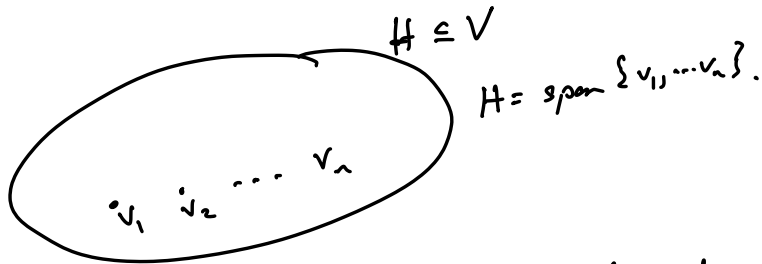
IMT says the columns are l.n. indep and span $\mathbb{R}^3$.

This is a basis for $\mathbb{R}^3$ which is not the standard basis.

Spanning set theorem

If V is a vector space and $H = \text{span}\{v_1, \dots, v_n\}$ then there is a subset of $\{v_1, \dots, v_n\}$ which is a basis for H .

Algorithm for finding a basis from a spanning set.



Step 1: Look at v_1 . If $v_1 = 0$ then discard it.
Otherwise keep it.

Step 2: Move on to v_2 . Assume $v_1 \neq 0$.

If $v_2 = kv_1$ for some k , then discard, otherwise keep it.

Step 3: Move on to v_3 . Assume v_1, v_2 weren't discarded. If $v_3 \in \text{span}\{v_1, v_2\}$ discard, otherwise keep it.

$\vdots$
Step k : We have kept $u_1, \dots, u_{k-1}$ up to now. We look v_k and we discard it if $v_k \in \text{span}\{u_1, \dots, u_{k-1}\}$. Otherwise we keep it.

$$\underbrace{v_1, v_2, v_3, v_4}_{\text{basis}}, v_5, v_6, v_7, \dots$$

Claim: This gives a basis for H .

Recall: If we have a matrix

$$A = \begin{pmatrix} | & & | \\ v_1 & \dots & v_k \\ | & & | \end{pmatrix} \quad \text{then we could be thinking}$$

of H as $\text{Col}(A)$ is $\text{span}\{v_1, \dots, v_k\}$.

$A \rightsquigarrow \text{rref}(A)$ and look at the columns of $\text{rref}(A)$ with pivots and the columns in A corr. to pivots form a basis.

$$\begin{pmatrix} 2 & 0 & 1 & 3 & 0 & 4 \\ 1 & 2 & 1 & 0 & 1 & 5 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix} = A$$



$$\begin{pmatrix} 1 & 0 & 0 & -\frac{3}{2} & -\frac{1}{2} & \frac{3}{2} \\ 0 & 1 & 0 & -\frac{3}{4} & \frac{1}{4} & \frac{5}{4} \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix} = \text{rref}(A)$$

$$\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \text{ is a basis for } \text{Col}(A).$$

Ex: A basis for the rowSpace of A.

$$A = \left(\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \\ 7 \end{pmatrix} \begin{pmatrix} 0 \\ 3 \\ 3 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} \right)$$

How do we find a basis for $\text{Row}(A)$?

$$\text{Row}(A) \subseteq \mathbb{R}^4.$$

Rowspace is not changed by
interchanging rows or mult.
a row by a non-zero constant.

Suppose r_i and r_j are both rows
and we replace $\underline{r_i}$ with $\underline{r_i + kr_j}$.

Does this change the rowspace? We can get r_i back
by computing $r_i + kr_j - kr_j$

$$A \rightsquigarrow \text{rref}(A) = \begin{pmatrix} 1 & 0 & 4.5 & -2.5 \\ 0 & 1 & -1.5 & 1.5 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The rows with pivots give a basis.

We also see that the first two columns of A form a basis for $\text{Col}(A)$

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 3 \\ 4 \\ 7 \end{pmatrix} \text{ is a basis.}$$

The size of a basis for $\text{Col}(A)$ = the size of a basis for $\text{Row}(A)$.

$$A^T = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 4 & 7 \\ 6 & 5 & 3 \\ 2 & 1 & 3 \end{pmatrix}$$

$$\text{rref}(A^T) = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 3 \\ 0 \\ 2 \end{pmatrix} \text{ and } \begin{pmatrix} 2 \\ 4 \\ 5 \\ 1 \end{pmatrix}$$

Form a basis
for $\text{Row}(A)$.

Basis for the columnspace

The pivot columns of a matrix A form a basis for the columnspace for A . That is, the columns in the rref of A with leading 1's correspond to the columns of A which form a basis for the columnspace of A .